

The rivalrous parts of non-rivalrous digital forms

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Peter Suber has asked, following a long chain of thinking about knowledge as a non-rivalrous form that is inscribed, historically, within rivalrous forms:

What is a public good? In the technical sense used by economists, a public good is non-rivalrous and non-excludable. A good is non-rivalrous when it's undiminished by consumption. We can all consume it without depleting it or becoming "rivals". Radio broadcasts are non-rivalrous; my reception doesn't block yours or vice versa. A good is non-excludable when consumption is available to all, and attempts to prevent consumption are generally ineffective. Radio broadcasts are non-excludable for people with the right equipment in the right area. Breathable air is non-excludable for this purpose even though a variety of barriers, from pollution to suffocation, could stop people from consuming it.

Knowledge is non-rivalrous. Your knowledge of a fact or idea does not block mine, and mine does not block yours. Thomas Jefferson described this situation beautifully in an 1813 letter to Isaac McPherson: “If nature has made any one thing less susceptible than all others of exclusive property, it is the action of the thinking power called an idea....Its peculiar character...is that no one possesses the less, because every other possesses the whole of it. He who receives an idea from me, receives instruction himself without lessening mine; as he who lights his taper at mine, receives light without darkening mine.” (See H.A. Washington, ed., The Writings of Thomas Jefferson, printed by the United States Congress, 1853-54, vol. VI, p. 180.)

Digital texts are often considered non-rivalrous. The dissemination of a digital text involves the first copy being produced (note the terminology here: even the first version is referred to as the first “copy”) and it can then be replicated without the original copy being destroyed or lost by the original owner.

Books and printed matter are not generally considered non-rivalrous since, even though the content inside can be copied and replicated, the actual means of transmission involve losing the original printed version.

Yet, I have been thinking about this. In the digital world, what does it mean to “produce a copy” and what are the material rivalrous underpinnings of that process?

Producing a copy over the internet (using HTTP over the TCP/IP protocol) involves, at a very basic level:

1. Both senders and recipients owning a computer and a monitor.
2. Both senders and recipients having an internet connection.

When the recipient (client) requests the document from the sender (server), the server retrieves the document from its hard drive, reads the contents into random access memory, and then serializes this into a stream of pulses down the various cablings of the internet. At the client end, the data are

received and saved into a temporary (or permanent) file having passed through memory. After decoding the file contents, the client's display is then instructed to illuminate the specific pixels that will render a humanly comprehensible version of the document.

Now, this is all somewhat trivial/obvious. However, the part that keeps bugging me is that while we call this a kind of non-rivalrous process, it relies upon a specific type of rivalrous, owned, technology that can shift its form of display dependent upon what is loaded into it. While the texts of which we speak are not rivalrous, they depend upon the rivalry for the computer display technology that can shift its form on command.

Is this a difference of *type* or of *degree* when compared to erasable tablets? I've thought of it as a difference of type until now. But it might, actually, be part of a far longer process. As Walter Benjamin put it: "In principle a work of art has always been reproducible. Man-made artifacts could always be imitated by men. [...] Around 1900 technical reproduction had reached a standard that not only permitted it to reproduce all transmitted works of art and thus to cause the most profound change in their impact upon the public; it also had captured a place of its own among the artistic processes."

Does the difference of degree reach an extreme point where we think of it as radically disruptive difference of type but it is still, really, only a difference of degree?

Suber never claimed that open access meant universal access. But I keep wondering how free we really are from those rivalrous rootings that underpinned the codex when we still require the display screen... Does it not still remain the case at the most basic level that, as Suber puts it of previous technologies, "we could [now: can] only record non-rivalrous knowledge in a rivalrous form"? Furthermore, are words actually knowledge in this case? What are the boundaries of representation vs the thing itself here and will it not always be the case that some rivalrous base will be needed to *represent* the workings of those abstract, non-rivalrous things that we call knowledge?